

Giovanni Paladino

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EDUCATION

New York University — PhD in Computer Science

Sept 2026 – Dec 2031

Arizona State University — BS in Computer Science — 4.0 GPA

May 2026

RESEARCH

ASTeRiSC Research Lab  — **Advised by Professor Adil Ahmad**

May 2025 – Current

IsolatedGaming (IsoGame) — **Under Review**

A systems solution for protecting games and players using secure VMs and performant graphics virtualization.

- Designed a **Linux** kernel driver in **C** for bi-directional memory shadowing to share **Vulkan** objects.
- Created **Vulkan** and **OpenGL** applications to intercept API calls for analyzing performance versus **virtio**.
- Present and review detailed technical talks and papers on systems security research in twice weekly meetings.

PROFESSIONAL

General Dynamics Mission Systems — **Software Engineer Intern**

May 2026 – Aug. 2026

- Built a **Rust** console to emulate serial communication between processes over **Linux** PTYs using **Tokio**.
- Developed a **Rust** command parser to automate end-to-end testing in a simulated hardware environment.
- Worked in a software engineering team to re-engineer legacy **C** code in idiomatic **Rust** to improve security.

Prescio Consulting — **Systems and Network Intern**

Oct 2022 – May 2026

- Provided network and server support on **Linux** and **Windows** environments for a remote engineering team.
- Worked with engineering team to design and build servers for Machine Learning and Data Science workloads.
- Utilized **Bash** scripting to diagnose network latency issues affecting remote engineers access to workstations.

PROJECTS

Enhanced Flashlight Glow  — **18,000+ Users and 99% Positive Ratings**

Dec 2024

- Leveraged a **Lua** modding API to improve lighting and enhance user accessibility in a popular Steam game.
- Continually iterated the software by quickly responding to user feedback and feature requests.
- Focused on correctness and performance by verifying effects with debuggers and using $O(1)$ data structures.

eBPF Vulkan Tracer 

June 2025

- Developed a custom **eBPF** tracer for understanding the **Linux** graphics pipeline by tracing a **Vulkan** program.
- Leveraged **Python** and **eBPF** to visualize the path of syscalls through a user space stack-trace in **Vulkan**.

TEACHING

Teaching Assistant — **Programming Languages**

Jan 2024 – May 2026

- Taught groups of 7+ computer science students **C** and **C++** programming twice a week in-person.
- Lectured on technical topics such as pointers, object references, and program memory.
- Supported students one-on-one, teaching coding and debugging in **C** and **C++** with **GDB** and **Valgrind**.

SKILLS

Languages and Libraries: C, C++, Rust, Python, Bash, Java, Lua, Javascript, OpenGL, Vulkan, eBPF

Tools, Databases, and OS: Neovim, Git, GDB, IDA, QEMU, LaTeX, Docker, PostgreSQL, Linux, Windows